

Perception in Flux: Investigating Memory and Attention During Gradual Environmental Transformations in Virtual Reality

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Figure 1: Different virtual environment configurations demonstrating possible settings.

ABSTRACT

This study examines how imperceptible environmental transitions in virtual reality (VR) influence spatial memory and perception. Participants experience a VR environment gradually morphing from one setting (a kitchen) to another (a bedroom) through the course of 15 minutes. We compare these morphing conditions to static controls to assess whether participants unconsciously form blended spatial memories integrating elements of both scenes. Eye-tracking data is used to analyze attentional shifts and their relationship to undetected changes. We hypothesize that participants in morphing conditions encode integrated spatial representations without consciously detecting transformations. This research aims to advance our understanding of how gradual transitions shape memory and perception, offering insights into designing immersive VR experiences that subtly adapt while maintaining presence.

Index Terms: Spatial Memory, Change Blindness, Eye Tracking, Cognitive Schemas

1 INTRODUCTION

Immersive environments provide a powerful platform to investigate how humans perceive, attend to, and remember spatial information. With VR, researchers can have precise control over environmental variables, surpassing the limitations of traditional laboratory tasks or two-dimensional displays. This makes it possible to study how subtle changes in our surroundings, often occurring without our conscious awareness, shape both our visual experience and our cognitive processes.

A prime example of these subtle phenomena is change blindness, which is the often-surprising failure to detect changes that would seem striking once noticed. While change blindness has been

documented extensively in static images and simple animations, its implications become increasingly complex and meaningful in spatially rich, dynamic VR scenarios. Here, transformations may happen gradually, allowing a scene to morph imperceptibly from one recognizable environment into another. Despite an observer's sense of continuous presence and engagement, the evolving environment may pass unnoticed, ultimately influencing how that individual encodes and later recalls the spatial layout.

1.1 Change Blindness in Immersive and Dynamic Environments

Foundational work by Simons and Rensink laid the theoretical groundwork for understanding change blindness, studying the interplay between perception, attention and memory, and showing that top-down cognitive schemas heavily guide what observers detect [17]. Research extending this foundation has revealed that change blindness is not an anomaly of attention under controlled conditions alone, but rather emerges even during ordinary viewing, affecting our comprehension of complex scenes [16].

This phenomenon retains its effects in immersive settings. Studies by Steinicke et al. explored how stereoscopic and monoscopic viewing conditions influence change detection, finding that depth cues do not necessarily improve one's ability to notice changes [18, 19]. Similarly, [13] examined change blindness in fully immersive 3D virtual environments, confirming that the phenomenon persists despite increased realism. This consistency does show the generality of change blindness: Be it flat or volumetric, users often fail to detect visual alterations when their attention is not directed toward the changes.

Building on these findings, several researchers have explored change blindness for practical applications. One study introduced "change blindness redirection," using unnoticed modifications to guide users through expansive virtual worlds within limited physical spaces [20]. Lohse et al. applied a similar logic to haptic remapping, rendering multiple virtual objects onto a single real-world prop without breaking user immersion [11]. In another pa-

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per, Hwang et al. further demonstrated that change blindness could facilitate redirected walking, enabling infinite navigation in finite rooms [6]. Such applications affirm that imperceptible changes can be exploited to create seamless, adaptive VR experiences.

Even subtle transformations have cognitive consequences. Levin et al. revealed a metacognitive bias, which they named “change blindness blindness”, where individuals overestimate their ability to spot alterations [10]. This finding suggests that slow, incremental modifications could be especially elusive. Frey et al. provided further insight, showing that gradual, slow-change stimuli are incorporated into memory without overt detection [2, 3]. This leaves observers with layered, blended representations of the scene.

1.2 Integrating Memory, Spatial Perception, and Imperceptible Changes

While change blindness research traditionally focused on detection rates, more recent studies have shifted toward understanding how undetected changes interact with spatial perception and memory. VR offers a playground where all spatial and visual parameters can be controlled and measured. One paper highlighted how VR can serve as a rigorous testbed for studying memory encoding and retrieval, particularly spatial and episodic memory [12]. These platforms enable the simulation of naturalistic scenes while systematically manipulating variables that shape memory performance.

Research on spatial fidelity and environmental realism has shown that high-fidelity VR scenes improve spatial memory accuracy [7]. Similarly, Vasser et al. noted that depth cues play a vital role in shaping spatial representations in 3D VR environments [22]. Work by Hollingworth and Henderson on incremental scene rotations suggest that gradual transformations may result in blended spatial memories rather than discrete recollections of an “original” or “final” state [5]. This blending is crucial for understanding how the human cognitive system integrates shifting cues over time. One study, focusing on saccade-contingent and blink-suppressed transformations in the context of hand redirection techniques [23], and another on examining saccade-driven updates [21], underscore the central role of attention and eye movements. Both studies show that subtle changes guided by ocular metrics can shape perceived continuity.

Gunesakaran et al. highlighted the potential of environmental cues in VR to enhance autobiographical recall, showing that carefully integrated spatial prompts can evoke vivid personal memories [4]. In another study, Banakou et al. explored the therapeutic use of imperceptible changes in VR environments to reduce anxiety. This work focused on desensitizing participants to anxiety-provoking scenarios by starting in a familiar, calming virtual environment that gradually transitioned to a less familiar or more stressful one [1]. By maintaining the subtlety of these transformations, participants were less likely to consciously detect the changes, which helped them adapt without triggering defensive reactions.

Furthermore, subtle manipulations of body and spatial cues in VR can affect self-representation. Şenel et al. demonstrated that imperceptible transformations to a virtual body can result in change blindness, particularly highlighting the saliency of self-representation and its impact on perception [15]. [14] and [9] add to this picture by emphasizing that attention, not just sensory input, dictates what is integrated into memory. These works jointly suggest that even unnoticed changes have cognitive consequences, shaping what we remember and how we navigate virtual worlds.

While previous research has established that

1. Change blindness is robust across a range of display types and immersive environments,
2. Gradual and imperceptible changes can be applied to redirect users or alter their experience of space, and

3. Subtle transformations influence memory and perception
- a key gap still remains.

Specifically, existing literature has not directly addressed how a seamless, slow morphing of an entire virtual environment—from one distinct space to another—affects integrated spatial memory representations. Previous work has examined changes to object positions, scene rotations, or body transformations individually. However, the question of whether participants can form stable, blended spatial memories when the entire environment gradually morphs from one familiar setting to another has not been comprehensively explored. This study addresses that gap. By using imperceptible transitions between two thematically distinct but structurally homomorphic VR environments, we investigate whether participants who are unaware of these changes still encode elements from both pre- and post-morph states. Additionally, we examine how top-down attentional processes, revealed through eye-tracking and gaze patterns, adapt to these transitions. In doing so, our research integrates and extends findings from change blindness, memory encoding in VR, attentional mechanisms, and environmental transformations. Ultimately, this work seeks to deepen our understanding of how gradual changes shape both the conscious and unconscious integration of spatial information in immersive virtual contexts.

2 METHODS

2.1 Experimental Design

Two structurally similar but thematically distinct virtual environments were created: a kitchen and a bedroom. These environments were designed with homomorphic mappings of key structural elements such as the walls, floors, and object placements, so that one can transition seamlessly into the other. Participants will be randomly assigned to one of four conditions:

- **Static Kitchen Condition:** Participants remain in an unchanging kitchen environment for an entire 15-minute session.
- **Static Bedroom Condition:** Participants remain in an unchanging bedroom environment for the entirety of the session.
- **Morphing Kitchen-to-Bedroom Condition:** Participants start in a kitchen environment that imperceptibly morphs into a bedroom over the course of the experiment.
- **Morphing Bedroom-to-Kitchen Condition:** Participants start in a bedroom environment that imperceptibly morphs into a kitchen over 15 minutes.

During the session, participants perform simple tasks, such as locating objects, tracing the objects through paths, and doing embodiment exercises. These tasks encourage participants to engage with the environment without drawing attention to potential changes.

2.2 Implementation

The virtual environments are implemented using Unity version 2022.3.15f1 LTS. Morphing transitions between environments are achieved using precomputed blend shapes for meshes, ensuring vertex consistency between corresponding objects. Mesh transitions are done using Unity Shader Graphs, where material properties such as base color and roughness are interpolated over time to produce seamless visual changes. For example, curtain colors are leaped from one state to another to reflect gradual environmental transformations. The Meta Quest Pro headset is used due to its integrated eye-tracking and hand-tracking capabilities. All transitions and interactions are controlled by a state machine that manages synchronization between participant tasks and environmental morphing.

For the trail-following task, participants interact with a red cube spawned on a table, which they move along a predefined path of spheres to a green target area (Figure 2). In the Qi Gong exercise, participants follow a semi-transparent sphere that dynamically moves around them, simulating flowing hand movements (Figure 3). These tasks synchronize with environmental changes to promote natural exploration without explicit attention to morphing.

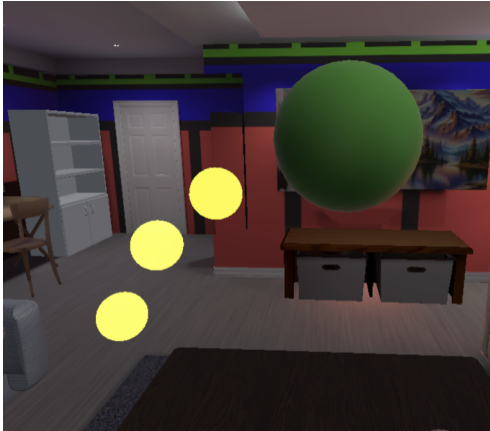


Figure 2: Trail exercise: Participants guide a red cube through a trail defined by spheres to a green target area.



Figure 3: Qi Gong exercise, where the user is instructed to follow the sphere's movements with their arms

2.3 Procedure

Before beginning the virtual session, participants complete a demographic questionnaire to ensure eligibility, and after this, they start the experiment. In the morphing conditions, changes to textures, object shapes, and structural elements begin immediately and unfold gradually throughout the session. Over the course of 15 minutes, participants interact naturally with the environment. The tasks are introduced via audio cues and auditory feedback is provided with each task completion. After the VR session, participants remove the headset and complete two sets of recall tasks:

- **Written Description:** They provide a free-form description of the environment they experienced.

- **Picture-Based Identification:** They select which reference images best match their recollection.

Finally, participants are directly asked whether they noticed any changes during the session.

2.4 Data Collection and Analysis

During the experiment, the following data are recorded and analyzed:

1. Behavioral Data

- **Memory Recall Accuracy:** We analyze written descriptions and picture-based identification results. Each participant's recall is classified as reflecting primarily the original environment, primarily the final environment, or a blended representation.
- **Change Detection:** Participants' explicit acknowledgment of environmental changes is recorded. We expect many participants in morphing conditions to remain unaware that any transformation occurred.

2. Eye-Tracking and Head-Gaze Data

- **Eye-Tracking Metrics:** Metrics such as fixation durations, saccade patterns, and dwell times are collected throughout the session to examine shifts in attention. These shifts provide insight into whether participants' patterns of attention adapt to the morphing environment, potentially reflecting top-down perceptual adjustments even without conscious detection.
- **Eye Scanpath Entropy:** The variability in participants' scan paths is analyzed to identify areas of visual saliency and how attention distribution aligns with environmental transformations, even without conscious detection. This analysis is informed by methods described in [8].

3. Qualitative Analysis of Spatial Representations

The qualitative content of the written descriptions is evaluated to determine if participants incorporate features from both the initial and final scenes. These analyses help pinpoint how gradual, unnoticed changes influence the internal representation of space.

2.5 Hypotheses

This is an exploratory study but our expectations are that:

1. **Blended Memory Representations:** Participants in the morphing conditions will recall elements from both the initial and final environments, resulting in blended spatial memories rather than clearly defined recollections of one stable scene.
2. **Low Change Detection Rates:** Despite these blended representations, most participants in the morphing conditions will not explicitly notice that the environment has changed.
3. **Top-Down Attentional Adaptation:** Eye-tracking data will reveal that participants' gaze patterns gradually shift to accommodate the emerging environment. This would support the notion that top-down processes guide attention in accordance with evolving contextual cues, even when the participant is not consciously aware of changes.

3 CONCLUSIONS

This study has been fully designed and implemented, though data collection has yet to begin. Once completed, we anticipate that the results will clarify how gradual, imperceptible transitions between distinct virtual environments influence spatial memory and attention. If participants form blended memories without noticing the changes, it would support the notion that incremental alterations are integrated unconsciously into cognitive representations. Likewise, if eye-tracking data reveal attentional shifts in step with the morphing environment, it would suggest that top-down perceptual processes adapt continuously to subtle contextual cues. Ultimately, this work stands to advance our understanding of how humans maintain a coherent sense of space as it evolves around them. Its findings may also inform practical applications, such as designing therapeutic VR scenarios that gently guide users from familiar to unfamiliar contexts, helping to create use cases for agoraphobia or creating adaptive virtual training environments that evolve to meet learners' needs. Through this research, we aim to provide insights into the cognitive underpinnings of perception, memory, and attention in dynamic, immersive virtual settings.

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